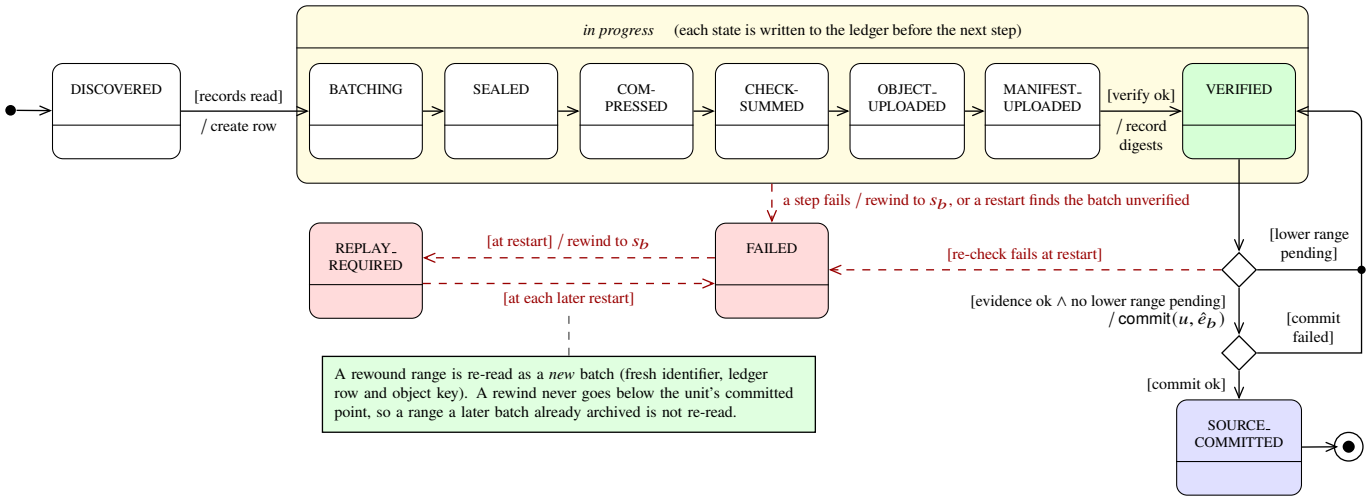

in progress (each state is written to the ledger before the next step)
DISCOVERED
[records read]
/ create row
BATCHING
SEALED
COM-
PRESSED
CHECK-
SUMMED
OBJECT_
UPLOADED
MANIFEST_
UPLOADED
[verify ok]
/ record
digests
VERIFIED
a step fails / rewind to s_b, or a restart finds the batch unverified
REPLAY_
REQUIRED
[at restart] / rewind to s_b
[at each later restart]
FAILED
[re-check fails at restart]
[lower range
pending]
[evidence ok ∧ no lower range pending]
/ commit(u, $\hat{e}_b$)
[commit
failed]
[commit ok]
SOURCE_
COMMITTED
A rewound range is re-read as a new batch (fresh identifier, ledger row and object key). A rewind never goes below the unit's committed point, so a range a later batch already archived is not re-read.